1. Customer Movie Search Page

The **Customer Movie Search Page** allows the customer to enter a movie or keyword and view available movies. The system retrieves movie and theater information from the **Movie & Theater Database**.

The database contains information such as movie titles, descriptions, ratings, theaters, showrooms, and available showtimes.

2. Movie/Showtime Selection Page

After selecting a movie, the customer goes to the **Movie/Showtime Selection Page**. This page displays available theaters, dates, and showtimes for the selected movie.

The page receives movie, theater, and showtime information from the **Movie & Theater Database**. The customer then selects the showing.

3. Seat Selection Page

After selecting a showtime, the customer is taken to the **Seat Selection Page**. This page displays the seating layout and current seat availability for the selected auditorium.

The system obtains seating information from the **Movie & Theater Database**. Seats can be available, temporarily held, or unavailable/sold. When a customer selects a seat, the system verifies that the seat is still available before continuing with the purchase.

Selected seats are held for **5 minutes** while the customer completes the payment process. If the customer does not complete the purchase within the time limit, the seats are released.

4. Login / Sign Up Page

The **Login / Sign Up Page** allows customers to create an account or log into an existing account.

The system stores customer information in the **Customer Database**, including account information, contact information, purchase history, and loyalty information. New accounts require a valid name, email address, and password. The system also prevents duplicate accounts and validates login information.

The architecture also connects authentication with CAPTCHA/bot protection

5. Payment Page

The **Payment Page** receives the customer's selected seat information and allows the customer to provide payment information.

The system supports external payment services such as credit cards, PayPal, and Bitcoin.

The Payment Page also sends the payment information to the **Bank / Payment Service**, which returns a payment status indicating whether the transaction was successful or failed.

If payment is successful, the system records the purchase and updates the customer's purchase history.

6. Bank / Payment Service

The **Bank / Payment Service** represents the external payment system used to process customer payments and refunds.

The Movie Theater Ticketing System does not directly process the customer's financial transaction. Instead, it communicates with an external payment service to validate and process the payment.

The payment processing and refunds require connections to external banking and payment services.

7. Customer Database

The **Customer Database** stores information related to customer accounts and purchases.

The database includes:

- Customer account information
- Contact information
- Payment-related information
- Purchase history
- Loyalty/membership information

After a successful purchase, the system updates the customer's purchase history and loyalty information when applicable.

8. Movie & Theater Database

The **Movie & Theater Database** stores information needed to allow customers to find and purchase tickets.

It contains:

- Movie information
- Movie descriptions
- Ratings
- Theater locations

- Showrooms/auditoriums
- Showtimes
- Seat layouts
- Seat availability

The application uses this database to display movies, showtimes, and real-time seat availability.

9. Confirmation Page

After the payment service confirms a successful transaction, the system sends the purchase information to the **Confirmation Page**.

The **Confirmation Page** displays the customer's successful booking and ticket information. This gives the customer immediate confirmation that the transaction was completed.

10. E-Ticket Delivery

The **E-Ticket Delivery** component generates and delivers the customer's electronic ticket after successful payment.

The system creates an electronic ticket and sends the ticket information to the customer.

If the payment is successful then generate E-tickets and provide them to the customer.

11. Email/SMS Notification Service

The **Email/SMS Notification Service** is an external service used to send booking-related messages to customers.

It can be used to send:

- Booking confirmations
- E-tickets
- Cancellation notifications